

LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Second-Hand Product Escrow Marketplace — Problem Statement #39

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Architecture Selection

The architecture selected for the Second-Hand Product Escrow Marketplace is Microservices Architecture.

Architectural Choice: The system is divided into independent components/services such as Order Manager, Payment Service, Inspection Service, Escrow Manager, and Dispute Manager. These components communicate through defined interfaces.

Reason 1: The marketplace contains separate responsibilities for ordering, payment, inspection, escrow, and dispute handling. Separating these responsibilities makes the system modular and allows individual components to be changed or maintained independently.

Reason 2: The escrow and dispute workflows have different processing needs. The architecture allows the relevant service to be handled independently, while the defined interfaces maintain communication between the services.

Security Advantage: Payment and escrow operations can be isolated within dedicated Payment Service and Escrow Manager components. This separation limits direct access to sensitive payment operations and allows security controls to be applied to the interfaces used for these operations.

Performance Benefit: Individual services can be scaled according to their workload. For example, payment or order-processing components can be scaled independently without requiring the entire marketplace system to be scaled together.

Components Identified for the Component Diagram

1. Order Manager 2. Payment Service 3. Inspection Service 4. Escrow Manager 5. Dispute Manager

Interfaces to Show

- Payment Processing API — Order Manager ↔ Payment Service
- Escrow Payment API — Payment Service ↔ Escrow Manager
- Inspection Report API — Inspection Service ↔ Order Manager
- Fund Freeze API — Dispute Manager ↔ Escrow Manager